

Intercity Bus Reservation System — Abstract

The Intercity Bus Reservation System is a software application designed to simplify and automate the management of bus schedules, passenger details, ticket bookings, and cancellation activities for long-distance bus travel. Traditional manual bus ticketing methods often rely on paper records, which can be time-consuming, error-prone, and difficult to maintain. This system provides a centralized platform for efficiently managing transport resources and passenger information.

The system allows administrators to add, update, search, and remove bus routes and maintain details such as bus ID, bus name, route source, destination, total seats, and ticket fare. It also manages passenger registration and maintains detailed booking records. Users can easily search for available buses on specific dates and view relevant route information. The system tracks seat availability in real time and helps operators identify fully booked or vacant buses, improving the efficiency of fleet scheduling.

By automating routine booking operations, the proposed system reduces paperwork, minimizes data-entry errors, saves time, and improves accessibility to transit information. It also provides an organized and reliable method for maintaining records and generating trip reports. The Intercity Bus Reservation System can be implemented as a web or desktop application and can be further enhanced with features such as automated email/SMS notifications, online payment integration, seat layout selection, and role-based access. Overall, the system aims to provide a simple, efficient, and user-friendly solution for managing intercity bus travel activities.